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# Experiment No.4 : Wi-Fi Network Scanning Using Mobile Apps (SSID, RSSI, Channel Mapping)

## Aim

To study and analyze nearby Wi-Fi networks using a mobile Wi-Fi analyzer application by identifying the **SSID, BSSID, RSSI, operating frequency, channel number, security type, and channel overlap**, and to understand their effect on wireless network performance.

## Theory

## Introduction

Wireless Local Area Networks (WLANs) based on the IEEE 802.11 standards have become the primary method of providing internet connectivity in homes, offices, educational institutions, and public places. Unlike wired networks, Wi-Fi uses **radio frequency (RF) signals** to transmit data between wireless devices and access points.

A Wi-Fi analyzer application available on Android or iOS devices can scan nearby wireless networks and display important parameters such as SSID, RSSI, channel number, frequency, security protocol, and signal strength. These parameters help network engineers optimize wireless network performance and troubleshoot connectivity issues.

This experiment introduces students to the practical analysis of Wi-Fi networks using mobile applications.

## Objectives

After performing this experiment, students will be able to:

1. Scan nearby Wi-Fi networks.
2. Identify SSID and BSSID.

3. Measure RSSI values.
4. Determine the operating frequency band.
5. Identify Wi-Fi channels.
6. Study channel overlap and interference.
7. Compare signal strengths of different networks.
8. Understand factors affecting Wi-Fi performance.

**Background Theory**

**1. What is Wi-Fi?**

Wi-Fi is a wireless networking technology that enables devices to communicate using radio waves.
It operates mainly in:

| Frequency Band | Range | Maximum Speed | Characteristics |
|---|---|---|---|
| **2.4 GHz** | Long | Lower | Better coverage but more interference |
| **5 GHz** | Medium | Higher | Faster with less interference |
| **6 GHz (Wi-Fi 6E/7)** | Short | Very High | Least interference and highest bandwidth |

**2. SSID (Service Set Identifier)**

SSID is the **name of the wireless network** that users see while connecting to Wi-Fi.
Examples:

- Home_WiFi
- College_Network
- Office_AP

Characteristics:

- Maximum length: 32 characters
- Can contain letters, numbers, and symbols
- Multiple routers may have different SSIDs.

Example:

```
SSID : Engineering Lab
```

### 3. BSSID (Basic Service Set Identifier)

BSSID is the **MAC address of the wireless access point**.

Example:

```
BSSID :
A4:CF:12:89:34:EF
```

Purpose:

- Identifies the physical wireless device.
- Two access points may have the same SSID but different BSSIDs.

### 4. RSSI (Received Signal Strength Indicator)

RSSI indicates how strong the received wireless signal is. It is measured in **dBm (decibel-milliwatts)**.

Since received power is much smaller than 1 mW, RSSI values are negative.

**Typical RSSI Values**

| RSSI | Signal Quality |
|---|---|
| -30 dBm | Excellent |
| -40 dBm | Very Strong |
| -50 dBm | Strong |
| -60 dBm | Good |
| -70 dBm | Fair |
| -80 dBm | Weak |
| -90 dBm | Very Weak |

Example:

```
RSSI = -45 dBm

Excellent signal
```

**5.Why is RSSI Negative?**

Wireless signals lose power while traveling through air due to:

- Distance
- Walls
- Furniture
- Human bodies
- Interference

Hence the received power is less than 1 mW and is expressed as negative dBm.

**6. Frequency Bands**

**2.4 GHz Band**

Advantages

- Longer range
- Better wall penetration

Disadvantages

- Heavy interference
- Limited channels

Frequency Range

2.400 GHz – 2.4835 GHz

**5 GHz Band**

Advantages

- Higher speed
- Less interference
- More channels

Disadvantages

- Shorter range

- **Why is RSSI Negative?**

  Wireless signals lose power while traveling through air due to:

  - Distance
  - Walls
  - Furniture
  - Human bodies
  - Interference

Hence the received power is less than 1 mW and is expressed as negative dBm.

- **Frequency Bands**

  **2.4 GHz Band**

  Advantages

  - Longer range
  - Better wall penetration

  Disadvantages

  - Heavy interference
  - Limited channels

  Frequency Range
  `2.400 GHz – 2.4835 GHz`

  **5 GHz Band**

  Advantages

  - Higher speed
  - Less interference
  - More channels

  Disadvantages

- Shorter range
- Helps select the best channel
- Reduces interference
- **Security Type**
  During scanning, the app also displays security information. Examples:
  - Open
  - WEP
  - WPA
  - WPA2
  - WPA3

  WPA2 and WPA3 are the most commonly recommended secure standards.

- **Mobile Wi-Fi Analyzer Apps**
  Examples include:
    - WiFi Analyzer
    - WiFiman
    - NetSpot
    - Network Analyzer

    These apps can display:
    - SSID
    - RSSI
    - Channel
    - Frequency
    - BSSID
    - Security type
    - Channel graph
    - Signal graph

**Apparatus Required**

- Android smartphone or iPhone
- Wi-Fi Analyzer application
- Multiple nearby Wi-Fi networks
- Laboratory notebook

**Procedure Step 1**
Install a Wi-Fi Analyzer application from the Play Store or App Store.

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**Step 2**
Enable:

- Wi-Fi
- Location permission (required for Wi-Fi scanning on many mobile operating systems)

**Step 3**
Open the Wi-Fi Analyzer application.

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**Step 4**
Allow the application to scan nearby wireless networks.

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**Step 5**

Observe the displayed information for each network:

- SSID
- BSSID
- RSSI
- Frequency
- Channel number
- Security protocol

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**Step 6**

Record the details of at least **10 nearby Wi-Fi networks**.

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**Step 7**
Open the **Channel Graph** or **Channel Rating** section in the application. Observe:

- Channel overlap
- Signal strengths
- Congested channels

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**Step 8**

Identify the least congested channel.

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**Step 9**
Compare networks operating in:

- 2.4 GHz
- 5 GHz

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**Step 10**
Analyze how RSSI changes when moving farther from the access point.

**Observation Table**

| Sr. No. | SSID | BSSID | Frequency (GHz) | Channel | RSSI (dBm) | Security | Band |
|---|---|---|---|---|---|---|---|
| 1 | vivo V29e | 3a:1e:4a:6c:95:a7 | 2.437 | 6 | -47 | WPA2 | 2.4 GHz |
| 2 | realme 8 5G | 7a:79:34:3f:17:7a | 2.462 | 11 | -49 | WPA2 | 2.4 GHz |
| 3 | OnePlus Nord 4 | 9a:0e:67:d5:c5:60 | 2.437 | 6 | -49 | WPA3 | 2.4 GHz |
| 4 | Redmi 9 | 1a:7a:2b:7a:b5:bc | 2.467 | 12 | -51 | WPA2 | 2.4 GHz |
| 5 | Galaxy A23 5G 886D | b2:63:09:e0:af:01 | 2.437 | 6 | -53 | WPA2 | 2.4 GHz |
| 6 | ameya | 96:22:31:ae:b5:58 | 2.437 | 6 | -53 | WPA2 | 2.4 GHz |


Access Points
2.4
5
6
GHz
PRO
IN RANGE 11
by Strength
vivo V29e
WiFi 4
3a:1e:4a:6c:95:a7 2m
2437MHz (20MHz) CH 6
WPA2
-47
dBm
realme 8 5G
WiFi 4
7a:79:34:3f:17:7a 3m
2462MHz (20MHz) CH 11
WPA2
-49
dBm
OnePlus Nord 4
WiFi 4
9a:0e:67:d5:c5:60 3m
2437MHz (20MHz) CH 6
WPA3
-49
dBm
Redmi 9
WiFi 4
1a:7a:2b:7a:b5:bc 3m
2467MHz (20MHz) CH 12
WPA2
-51
dBm
Galaxy A23 5G 886D
WiFi 4
b2:63:09:e0:af:01 4m
2437MHz (20MHz) CH 6
WPA2
-53
dBm
ameya
WiFi 4
96:22:31:ae:b5:58 4m
2437MHz (20MHz) CH 6
WPA2
-53
dBm

**Result Analysis**

Students should analyze:

- Which network has the strongest RSSI?
- Which channels are most crowded?
- Which frequency band has more networks?
- Are non-overlapping channels (1, 6, 11) being used?
- Which network is likely to provide better performance?

Based on the Wi-Fi networks visible in your screenshot:

1. **Which network has the strongest RSSI?**
   The **vivo V29e** network has the strongest RSSI of **-47 dBm**. A less negative RSSI indicates a stronger received signal.
2. **Which channels are most crowded?**
   **Channel 6** is the most crowded in the visible networks, with **four networks** using it:
   - vivo V29e
   - OnePlus Nord 4
   - Galaxy A23 5G 886D
   - ameya
3. Channel 11 has two visible networks, while Channel 12 has one.
4. **Which frequency band has more networks?**
   In the screenshot provided, all **7 visible networks are operating in the 2.4 GHz band**. Therefore, based on this screenshot, **2.4 GHz has more networks**. A complete comparison with 5 GHz requires the 5 GHz scan.
5. **Are non-overlapping channels (1, 6, 11) being used?**
   Yes. **Channel 6 and Channel 11** are being used. **Channel 1 is not visible** in the current scan. The experiment identifies channels 1, 6, and 11 as the recommended non-overlapping 2.4 GHz channels.

6. **Which network is likely to provide better performance?**
   Based **only on RSSI**, **vivo V29e** is likely to provide the better signal among the visible networks because it has the strongest RSSI at **-47 dBm**. However, actual Wi-Fi performance also depends on channel congestion, interference, distance, and other factors. Your experiment specifically asks you to consider these factors when analyzing performance.

**Precautions**

1. Enable Wi-Fi before scanning.
2. Grant location permission to the application.
3. Do not modify router settings during the experiment.
4. Perform scanning from the same location for consistent readings.
5. Record RSSI values carefully.
6. Avoid moving while taking measurements.

**Applications**

- Wi-Fi network planning
- Site surveys
- Wireless troubleshooting
- Network optimization
- Access point placement
- Coverage analysis
- Smart home network setup
- Enterprise WLAN management

**Advantages**

- Easy to analyze wireless networks
- Helps detect interference
- Identifies weak signal areas
- Assists in selecting the best Wi-Fi channel
- Improves network performance

## Conclusion

In this experiment, students used a mobile Wi-Fi analyzer application to scan nearby wireless networks and examine key parameters such as **SSID, BSSID, RSSI, frequency band, channel number, and security type**. By analyzing channel mapping and signal strength, they observed how **channel overlap and interference** affect Wi-Fi performance. This practical provides a foundation for understanding wireless network planning, optimization, and troubleshooting in real-world environments.